

Project 31

Hume City Council CyberSafe Services – Online Scam Detection and Reporting System

About the Client

Hume City Council is one of Australia's fastest-growing municipalities, located 20 kilometres north of Melbourne's CBD. It covers both established urban areas in the south and rural landscapes in the north, serving a diverse community of over 262,000 residents. Council headquarters are located at 1079 Pascoe Vale Road, Broadmeadows.

The municipality is a major economic and transport hub encompassing the Melbourne Airport, the Hume and Calder Highways, and the Western Ring Road.

Key services and facts about Hume City Council include:

- **Community Services:** Council provides a wide range of support for residents, including maternal and child health, immunisations, youth assertive outreach (ages 10-24), aged care, and disability support.
- **Environment & Sustainability:** Managing over 160,000 trees, the council leads initiatives like Hume Home Energy Help, climate action planning, and Green Wedge protection.
- **Advocacy & Growth:** As a fast-growing interface council, the local government actively advocates for State and Federal investment in key infrastructure, transport, and community safety projects.

Project Brief:

Online Cybersecurity Assessment System is an online cybersecurity assessment portal designed to help small businesses, community organizations and not-for-profit organizations identify cybersecurity risks and improve their security practices.

The system will allow organizations to complete cybersecurity assessments, receive risk scores, view recommendations, monitor improvements and access cybersecurity awareness resources.

System Features

In the life of the software development, problem analysis provides a base for design and development phase. The problem is analyzed so that sufficient matter is provided to design a new system. Large problems are sub-divided into smaller ones to make them understandable and easy for finding solutions. Same in this project all the tasks are sub-divided and categorized.

System Modules:

- User registration and authentication module
- User profile and privacy module
- Scam checking and analysis module
- URL and contact indicator analysis module
- Scam reporting module
- Evidence management module
- Report review and verification module
- Community scam alert module
- Scam map and trend-analysis module
- Scam awareness and education module
- Notification and subscription module
- Support and recovery guidance module
- Administration and configuration module
- Reporting, analytics and audit module

Functional Requirements

Functional requirements are product features or functions that developers must implement to enable users to accomplish their tasks. So, it's important to make them clear for the stakeholders. Generally, functional requirements describe system behavior under specific conditions. The developers of this system must enhance the performance and efficiency of the system by adding 15 to 20 more functional requirements. Students need to do their own research to find how they can improve the system and which FRs need to be added. The group must need a prior approval from the stakeholders/project supervisor before finalizing these Functional Requirements. These enhanced FRs must be reflected separately in Final SRS Report after the approval.

Module 1: User Registration and Authentication

FR1: User registration

FR2: Email verification

FR3: Duplicate email validation

FR4: User login

FR5: User logout

FR6: Password recovery

Module 2: User Profile and Access Control

FR7: Profile viewing

FR8: Profile updating

FR9: Password changing

FR10: Role-based access control

FR11: Notification preference management

FR12: Account deletion request

Module 3: Scam Content Analysis

FR13: Suspicious text submission

FR14: Communication type selection

FR15: Scam keyword detection

FR16: Scam risk score calculation

FR17: Scam risk classification

FR18: Scam indicator display

Module 4: URL and Contact Analysis

- FR19:** Suspicious URL submission
- FR20:** URL format validation
- FR21:** Suspicious URL pattern detection
- FR22:** Suspicious phone number submission
- FR23:** Suspicious email address submission
- FR24:** Reported indicator matching

Module 5: Scam Report Management

- FR25:** Scam report creation
- FR26:** Draft report saving
- FR27:** Report field validation
- FR28:** Report reference generation
- FR29:** Submitted report viewing
- FR30:** Report withdrawal

Module 6: Evidence Management

- FR31:** Evidence file uploading
- FR32:** File type validation
- FR33:** File size validation
- FR34:** Evidence description entry
- FR35:** Evidence access restriction
- FR36:** Evidence access logging

Module 7: Report Review and Verification

- FR37:** Review queue display
- FR38:** Reviewer assignment
- FR39:** Report classification
- FR40:** Report severity updating
- FR41:** Additional information request
- FR42:** Report approval and rejection

Module 8: Duplicate and Related Report Detection

- FR43:** Duplicate report detection
- FR44:** Report text similarity comparison
- FR45:** Phone number comparison
- FR46:** Email address comparison

FR47: Website address comparison

FR48: Related report linking

Module 9: Community Scam Alerts

FR49: Scam alert creation

FR50: Reporter information anonymisation

FR51: Scam alert approval

FR52: Scam alert publication

FR53: Scam alert searching

FR54: Scam alert archiving

Module 10: Scam Awareness and Recovery

FR55: Scam awareness resource viewing

FR56: Resource categorisation

FR57: Resource searching

FR58: Scam response checklist display

FR59: Resource recommendation

FR60: Recovery action tracking

Module 11: Notification and Subscription

FR61: Report submission notification

FR62: Report status notification

FR63: Information request notification

FR64: Scam category subscription

FR65: Regional alert subscription

FR66: Notification unsubscription

Module 12: Administration, Reporting and Audit

FR67: User account management

FR68: User role management

FR69: Scam category management

FR70: Scam report statistics generation

FR71: De-identified data export

FR72: User activity logging

Non-Functional Requirements (NFRs)

Performance

1. The system should load all pages within 3 seconds under normal network conditions.
2. The system should support at least 1000 concurrent users.
3. The database should respond to queries in less than 2 seconds for up to 100,000 records.

Reliability

4. The system should maintain 99% uptime excluding scheduled maintenance.
5. The system should provide automatic backup of user data daily.
6. In case of system failure, the database should restore from the latest backup without data loss exceeding 24 hours.

Security

7. The system should use HTTPS (SSL/TLS) for secure data transmission.
8. Passwords and sensitive information should be encrypted and never stored in plaintext.
9. The system should prevent SQL Injection, XSS, and CSRF attacks.
10. User sessions should automatically expire after inactivity.

Usability

11. The interface should be intuitive and mobile-responsive.
12. Form validations should display clear error messages.
13. The design should use consistent layouts and colour themes.
14. The system should support both desktop and mobile browsers.

Maintainability

15. The system should follow modular PHP code structure with MVC pattern.

- 16.All functions should be documented for future maintenance.
- 17.The database schema should be normalised up to 3NF for scalability.

Scalability

- 18.The system should support easy migration from single-server MySQL to clustered database architecture.
- 19.The design should allow integration with REST APIs for mobile app extension.

Availability

- 20.The system should be available 24 × 7 with scheduled downtime notifications.
- 21.The system should automatically recover from unexpected server shutdowns using error logs.

Data Integrity

- 22.The system should ensure consistency between linked tables (foreign-key constraints).
- 23.The system should validate all input before committing to the database.

Compliance

- 24.The system should comply with GDPR-like data privacy standards, allowing users to delete their profiles permanently.
- 25.All personal information should be used only for matchmaking purposes.

Ethical Requirements

The system should avoid creating fear or presenting assessment results as professional cybersecurity certification.

The system should clearly state that:

- Results are based on information provided by the organisation
- The assessment provides general guidance
- The portal does not guarantee protection from cyberattacks
- Serious cybersecurity incidents may require assistance from qualified professionals or relevant authorities

The scoring process shall be transparent and shall not unfairly discriminate between organisations based on their size, industry or available resources.

Hardware Requirement: Should be recommended by the developers.

Software Requirement: Should be recommended by the developers.